NBA Total Points Prediction: Predictive Modeling

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Library Loading

```r
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```r
library(ggplot2)
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.2

## Loading required package: lattice
```

```r
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.5.2

## Loading required package: Matrix

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

## Loaded glmnet 4.1-10
```

```r
library(rpart)
library(car)
```

```
## Warning: package 'car' was built under R version 4.5.2

## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode
```

```r
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```r
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.5.2

## randomForest 4.7-1.2

## Type rfNews() to see new features/changes/bug fixes.

##
## Attaching package: 'randomForest'

## The following object is masked from 'package:ggplot2':
##
##     margin

## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
library(gbm)
```

```
## Warning: package 'gbm' was built under R version 4.5.2

## Loaded gbm 2.2.3

## This version of gbm is no longer under development. Consider transitioning to gbm3, https://github.c
```

```r
library(nnet)
library(Metrics)
```

```
## Warning: package 'Metrics' was built under R version 4.5.2

##
## Attaching package: 'Metrics'

## The following objects are masked from 'package:caret':
##
##     precision, recall
```

```r
library(tibble)
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.2

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 4.5.2

##
## Attaching package: 'purrr'

## The following object is masked from 'package:car':
##
##     some

## The following object is masked from 'package:caret':
##
##     lift
```

```r
library(e1071)
```

```
## Warning: package 'e1071' was built under R version 4.5.2

##
## Attaching package: 'e1071'

## The following object is masked from 'package:ggplot2':
##
##     element
```

```r
library(rprojroot)
```

```
## Warning: package 'rprojroot' was built under R version 4.5.2
```

```r
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.5.2

## corrplot 0.95 loaded
```

```r
model_ready <- read_csv("Data/FINAL_MODEL_READY.csv", show_col_types = FALSE) |>
  mutate(game_date = as.Date(game_date)) |>
  filter(
    game_date >= as.Date("2013-10-01"),
    month(game_date) %in% c(10, 11, 12, 1, 2, 3, 4)
  )

cat("Duplicate game_ids:", sum(duplicated(model_ready$game_id)), "\n")
```

```
## Duplicate game_ids: 0
```

```r
cat("Total rows:", nrow(model_ready), "\n")
```

```
## Total rows: 15702
```

```r
cat("Date range:", as.character(min(model_ready$game_date)),
    "to", as.character(max(model_ready$game_date)), "\n")
```

```
## Date range: 2013-10-18 to 2026-03-03
```

TP Dist.

```r
summary(model_ready$total)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   134.0   202.0   217.0   217.5   233.0   351.0
```

```r
cat("Skewness:", round(e1071::skewness(model_ready$total), 3), "\n")
```

```
## Skewness: 0.139
```

```r
cat("Kurtosis:", round(e1071::kurtosis(model_ready$total), 3), "\n")
```

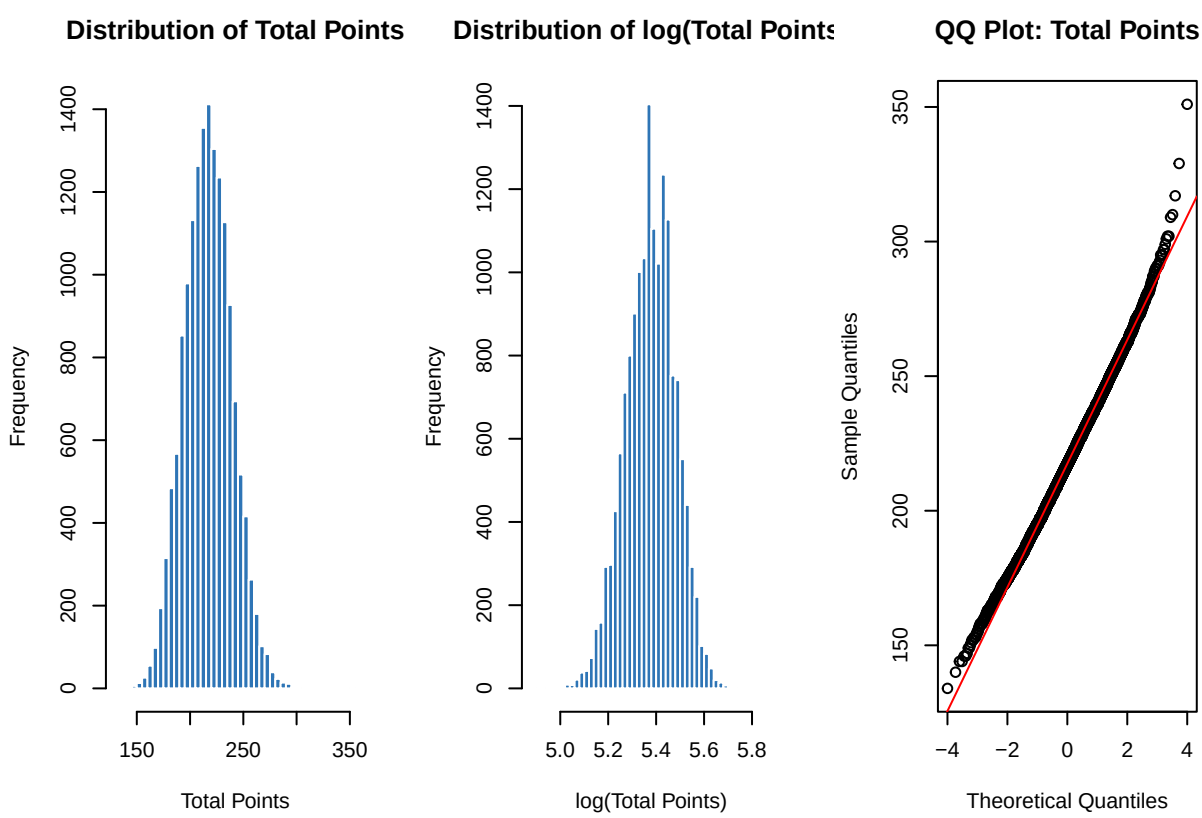
```
## Kurtosis: 0.088
```

```r
par(mfrow = c(1, 3))

hist(model_ready$total,
     breaks = 50,
     main   = "Distribution of Total Points",
     xlab   = "Total Points",
     col    = "#2E75B6", border = "white")

hist(log(model_ready$total),
     breaks = 50,
     main   = "Distribution of log(Total Points)",
     xlab   = "log(Total Points)",
     col    = "#2E75B6", border = "white")

qqnorm(model_ready$total, main = "QQ Plot: Total Points")
qqline(model_ready$total, col = "red")
```



```r
par(mfrow = c(1, 1))
```

Train and testing data

```r
cutoff_date <- as.Date("2025-03-13")

train <- model_ready |> filter(game_date <  cutoff_date)
test  <- model_ready|> filter(game_date >= cutoff_date)

cat("Training set:", nrow(train), "games (",
    as.character(min(train$game_date)), "to",
    as.character(max(train$game_date)), ")\n")
```

```
## Training set: 14493 games ( 2013-10-18 to 2025-03-12 )
```

```r
cat("Test set:    ", nrow(test),  "games (",
    as.character(min(test$game_date)),  "to",
    as.character(max(test$game_date)),  ")\n")
```

```
## Test set:     1209 games ( 2025-03-13 to 2026-03-03 )
```

Feature matrix

```r
id_cols <- c("game_id", "game_date", "home_team", "away_team", "total", "oreb", "spread")

numeric_data <- model_ready |>
  select(where(is.numeric)) |>
  select(-any_of(c("game_id", "total", "oreb","spread")))

# remove zero-variance columns first
zero_var <- names(numeric_data)[sapply(
  numeric_data,
  function(x) var(x, na.rm = TRUE) == 0 | all(is.na(x))
)]

if (length(zero_var) > 0) {
  cat("Dropping zero-variance columns:\n")
  print(zero_var)
  numeric_data <- numeric_data |>
    select(-all_of(zero_var))
}

# remove obvious collinear / redundant variables before VIF
drop_for_vif <- c(
  "blend_total_proj5",
  "rebound_matchup_edge",
  "orb_pg_diff",
  "drb_pg_diff",
```

```r
  "trb_pg_diff",
  "x3pa_rate_pg_diff",
  "shooting_env5",
  "pace_pressure5",
  "free_throw_env5",
  "tov_env5",
  "pace_rebound_pressure"
)

vif_vars <- setdiff(names(numeric_data), drop_for_vif)

lm_test <- lm(
  total ~ .,
  data = model_ready |>
    select(total, all_of(vif_vars))
)

aliased <- alias(lm_test)$Complete
if (!is.null(aliased)) {
  cat("Aliased coefficients detected. Remove some of these variables:\n")
  print(aliased)
}
```

```
## Aliased coefficients detected. Remove some of these variables:
##                     (Intercept) season opp_pts_allowed_diff opp_orb_allowed_diff
## def_rtg_diff         0           0      0                     0
## net_rtg_diff         0           0      0                     0
## last5_net_rtg_diff   0           0      0                     0
##                     opp_trb_allowed_diff opp_ast_allowed_diff
## def_rtg_diff         0                    0
## net_rtg_diff         0                    0
## last5_net_rtg_diff   0                    0
##                     opp_tov_forced_diff off_rtg_diff efg_pct_diff tov_pct_diff
## def_rtg_diff         0                  -1            0            0
## net_rtg_diff         0                   2            0            0
## last5_net_rtg_diff   0                   0            0            0
##                     orb_pct_diff ft_rate_diff last5_off_rtg_diff
## def_rtg_diff         0            0            0
## net_rtg_diff         0            0            0
## last5_net_rtg_diff   0            0            1
##                     last5_def_rtg_diff last5_pace_diff last5_pts_diff
## def_rtg_diff         0                  0               0
## net_rtg_diff         0                  0               0
## last5_net_rtg_diff  -1                  0               0
##                     last5_pts_allowed_diff last5_oreb_diff rest_diff
## def_rtg_diff         0                      0               0
## net_rtg_diff         0                      0               0
## last5_net_rtg_diff   0                      0               0
##                     home_back_to_back away_back_to_back missed_fg_pg_diff
## def_rtg_diff         0                 0                 0
## net_rtg_diff         0                 0                 0
## last5_net_rtg_diff   0                 0                 0
##                     expected_missed_fg orb_per100_diff drb_per100_diff
```

```
## def_rtg_diff        0                  0               0
## net_rtg_diff        0                  0               0
## last5_net_rtg_diff  0                  0               0
##                    trb_per100_diff x3pa_rate_per100_diff last5_missed_fg_diff
## def_rtg_diff        0               0                     0
## net_rtg_diff        0               0                     0
## last5_net_rtg_diff  0               0                     0
##                    last10_missed_fg_diff last5_missed_3pt_diff
## def_rtg_diff        0                     0
## net_rtg_diff        0                     0
## last5_net_rtg_diff  0                     0
##                    last10_missed_3pt_diff last5_x3pa_rate_diff
## def_rtg_diff        0                      0
## net_rtg_diff        0                      0
## last5_net_rtg_diff  0                      0
##                    last10_x3pa_rate_diff last10_oreb_diff
## def_rtg_diff        0                     0
## net_rtg_diff        0                     0
## last5_net_rtg_diff  0                     0
##                    last5_oreb_allowed_diff last10_oreb_allowed_diff
## def_rtg_diff        0                       0
## net_rtg_diff        0                       0
## last5_net_rtg_diff  0                       0
##                    expected_oreb_pressure home_oreb_matchup away_oreb_matchup
## def_rtg_diff        0                      0                 0
## net_rtg_diff        0                      0                 0
## last5_net_rtg_diff  0                      0                 0
```

```r
keep_coef <- names(coef(lm_test))[!is.na(coef(lm_test))]
keep_coef <- setdiff(keep_coef, "(Intercept)")

lm_test_vif <- lm(
  as.formula(paste("total ~", paste(keep_coef, collapse = " + "))),
  data = model_ready |>
    select(total, all_of(keep_coef))
)

cat("VIF values:\n")
```

```
## VIF values:
```

```r
print(car::vif(lm_test_vif))
```

```
##                   season     opp_pts_allowed_diff     opp_orb_allowed_diff
##                 1.144776                 4.970503                 4.384938
##     opp_trb_allowed_diff     opp_ast_allowed_diff      opp_tov_forced_diff
##                13.505144                 2.091688                 2.729061
##             off_rtg_diff             efg_pct_diff             tov_pct_diff
##                23.818498                22.732706                 5.471013
##             orb_pct_diff             ft_rate_diff       last5_off_rtg_diff
##                 3.654795                 1.738106               151.798527
##       last5_def_rtg_diff          last5_pace_diff           last5_pts_diff
##               142.518554               175.622410               187.986335
```

```
##   last5_pts_allowed_diff          last5_oreb_diff                rest_diff
##               190.508535                 4.201764                 1.039768
##        home_back_to_back        away_back_to_back        missed_fg_pg_diff
##                 1.047407                 1.064438                13.830063
##       expected_missed_fg          orb_per100_diff          drb_per100_diff
##                 1.503307               589.686357               761.309561
##          trb_per100_diff    x3pa_rate_per100_diff     last5_missed_fg_diff
##              1283.542959                 4.008337                 7.794995
##    last10_missed_fg_diff    last5_missed_3pt_diff   last10_missed_3pt_diff
##                 8.382095                20.300995                30.191295
##     last5_x3pa_rate_diff    last10_x3pa_rate_diff         last10_oreb_diff
##                20.744733                31.449913                 4.919304
##  last5_oreb_allowed_diff last10_oreb_allowed_diff   expected_oreb_pressure
##                64.441198                 2.655134                 1.994835
##        home_oreb_matchup        away_oreb_matchup
##                54.641153                54.131018
```

```r
# OLS-friendly feature set
feat_cols_ols <- setdiff(names(model_ready), id_cols)

feat_drop_ols <- c(
  "rebound_matchup_edge",
  "orb_pg_diff",
  "drb_pg_diff",
  "trb_pg_diff",
  "x3pa_rate_pg_diff",
  "orb_per100_diff",
  "drb_per100_diff",
  "trb_per100_diff",
  "net_rtg_diff",
  "last5_net_rtg_diff",
  "blend_total_proj5"
)

feat_cols_ols <- setdiff(feat_cols_ols, feat_drop_ols)

# richer tree-model feature set
feat_cols_gbm <- setdiff(names(model_ready), id_cols)

cat("Number of OLS predictors:", length(feat_cols_ols), "\n")
```

```
## Number of OLS predictors: 41
```

```r
cat("Number of GBM predictors:", length(feat_cols_gbm), "\n")
```

```
## Number of GBM predictors: 52
```

```r
X_train_ols <- train[, feat_cols_ols, drop = FALSE]
X_test_ols  <- test[, feat_cols_ols, drop = FALSE]

X_train_gbm <- train[, feat_cols_gbm, drop = FALSE]
X_test_gbm  <- test[, feat_cols_gbm, drop = FALSE]
```

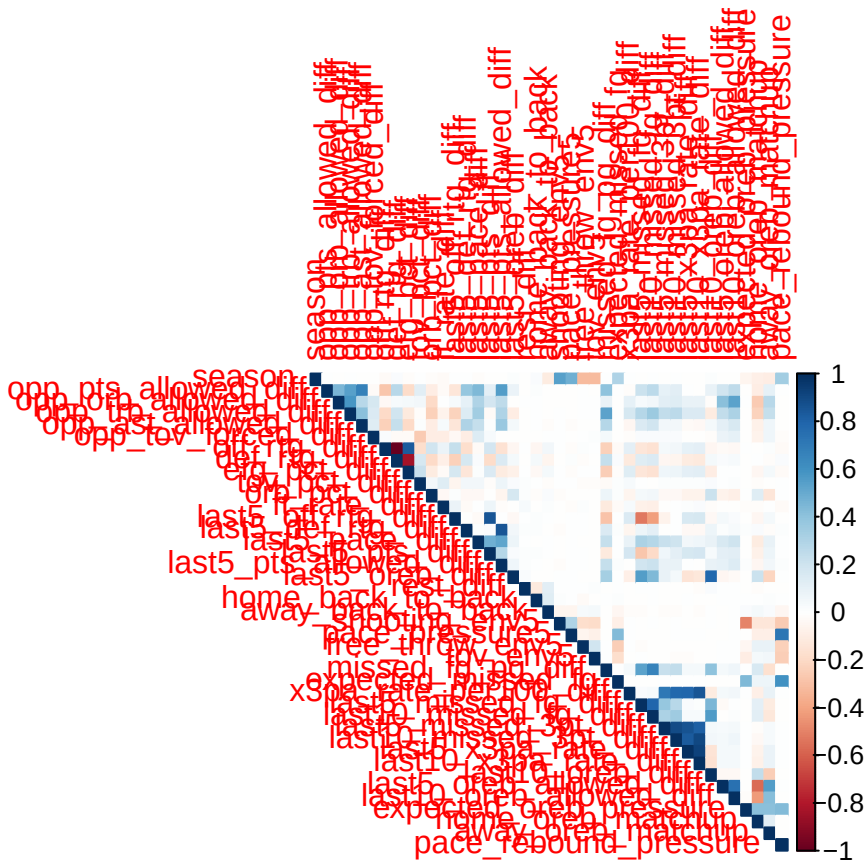
```r
y_train <- train$total
y_test  <- test$total

y_train_log <- log(y_train)
y_test_log  <- log(y_test)

na_counts <- colSums(is.na(X_train_ols))
if (any(na_counts > 0)) {
  cat("WARNING - NAs in training features:\n")
  print(na_counts[na_counts > 0])
} else {
  cat("No NAs in training feature matrix.\n")
}
```

```
## WARNING - NAs in training features:
##            shooting_env5           pace_pressure5          free_throw_env5
##                        5                        5                        5
##                 tov_env5     last5_missed_fg_diff    last10_missed_fg_diff
##                        5                        5                       10
##    last5_missed_3pt_diff   last10_missed_3pt_diff     last5_x3pa_rate_diff
##                        5                       10                        5
##    last10_x3pa_rate_diff         last10_oreb_diff  last5_oreb_allowed_diff
##                       10                       10                        5
## last10_oreb_allowed_diff   expected_oreb_pressure        home_oreb_matchup
##                       10                        5                        2
##        away_oreb_matchup    pace_rebound_pressure
##                        3                        5
```

```r
cor_matrix <- cor(X_train_ols |> select(where(is.numeric)), use = "complete.obs")
corrplot(cor_matrix, method = "color", type = "upper")
```




```r
cor_with_total <- sapply(
  X_train_ols |> select(where(is.numeric)),
  function(x) cor(y_train, x, use = "complete.obs")
)

cat("Top correlations with total:\n")
```

```
## Top correlations with total:
```

```r
print(sort(abs(cor_with_total), decreasing = TRUE))
```

```
##           pace_pressure5                   season            shooting_env5
##             0.4347192142             0.4296835587             0.3425872400
##    pace_rebound_pressure       expected_missed_fg                 tov_env5
##             0.2700731502             0.2064992128             0.1872126902
##          free_throw_env5   expected_oreb_pressure        away_back_to_back
##             0.0904294992             0.0525572728             0.0510715461
##        away_oreb_matchup        home_oreb_matchup     last5_missed_fg_diff
##             0.0357747939             0.0287789276             0.0217623043
##             tov_pct_diff            off_rtg_diff             def_rtg_diff
##             0.0179163555             0.0177122171             0.0177122171
##     opp_orb_allowed_diff          last5_pace_diff        missed_fg_pg_diff
##             0.0134635331             0.0129029426             0.0126530243
##  last5_oreb_allowed_diff             orb_pct_diff    last10_missed_fg_diff
```

```
##             0.0121202316             0.0108863060             0.0105301253
##     opp_trb_allowed_diff             ft_rate_diff last10_oreb_allowed_diff
##             0.0097977018             0.0092953997             0.0092116356
##       last5_off_rtg_diff         last10_oreb_diff      opp_tov_forced_diff
##             0.0086738332             0.0071297866             0.0065916932
##       last5_def_rtg_diff    last5_missed_3pt_diff          last5_oreb_diff
##             0.0065430307             0.0063865707             0.0058087124
##                rest_diff     opp_ast_allowed_diff             efg_pct_diff
##             0.0033009755             0.0032645896             0.0032463111
##     opp_pts_allowed_diff   last10_missed_3pt_diff        home_back_to_back
##             0.0028630697             0.0028516027             0.0015260771
##    last10_x3pa_rate_diff           last5_pts_diff   last5_pts_allowed_diff
##             0.0014536391             0.0005881399             0.0004282906
##     last5_x3pa_rate_diff    x3pa_rate_per100_diff
##             0.0003130274             0.0002457291
```

Cross-validation

```r
ts_ctrl <- trainControl(
  method          = "timeslice",
  initialWindow   = floor(nrow(train) * 0.6),
  horizon         = floor(nrow(train) * 0.08),
  fixedWindow     = FALSE,
  summaryFunction = defaultSummary,
  allowParallel   = TRUE
)

cv5_ctrl <- trainControl(
  method   = "cv",
  number   = 5,
  summaryFunction = defaultSummary
)

mae_raw <- function(actual, predicted) {
  mean(abs(actual - predicted), na.rm = TRUE)
}

mae_log_backtransform <- function(actual, pred_log) {
  mean(abs(actual - exp(pred_log)))
}
```

Baseline Model

```r
baseline_pred <- rep(mean(y_train), length(y_test))
baseline_mae  <- mae_raw(y_test, baseline_pred)

cat("Baseline (training mean prediction):", round(mean(y_train), 2), "points\n")
```

```
## Baseline (training mean prediction): 216.53 points
```

```r
cat("Baseline MAE on test set:", round(baseline_mae, 2), "\n")
```

```
## Baseline MAE on test set: 18.84
```

Model 1: Ordinary Least Squares Linear Regression

o l s l o l

Full Model 1

```r
lm_full <- lm(total ~ ., data = cbind(total = y_train, X_train_ols))
lm_full_pred <- predict(lm_full, newdata = X_test_ols)
lm_full_mae <- mae_raw(y_test, lm_full_pred)

cat("OLS Full Model - Test MAE:", round(lm_full_mae, 2), "\n")
```

```
## OLS Full Model - Test MAE: 15.61
```

```r
cat("R-squared (training):", round(summary(lm_full)$r.squared, 4), "\n")
```

```
## R-squared (training): 0.277
```

```r
if (requireNamespace("car", quietly = TRUE)) {
  aliased_full <- alias(lm_full)$Complete

  if (!is.null(aliased_full)) {
    cat("\nAliased coefficients detected in full OLS model.\n")
    cat("VIF cannot be computed until redundant predictors are removed.\n")
    print(aliased_full)
  } else {
    vif_vals <- car::vif(lm_full)
    cat("\nPredictors with VIF > 10 (multicollinearity concern):\n")
    print(vif_vals[vif_vals > 10])
  }
}
```

```
##
## Aliased coefficients detected in full OLS model.
## VIF cannot be computed until redundant predictors are removed.
##              (Intercept) season opp_pts_allowed_diff opp_orb_allowed_diff
## def_rtg_diff  0           0      0                    0
##              opp_trb_allowed_diff opp_ast_allowed_diff opp_tov_forced_diff
## def_rtg_diff  0                    0                    0
##              off_rtg_diff efg_pct_diff tov_pct_diff orb_pct_diff ft_rate_diff
## def_rtg_diff -1            0            0            0            0
##              last5_off_rtg_diff last5_def_rtg_diff last5_pace_diff
## def_rtg_diff  0                  0                  0
##              last5_pts_diff last5_pts_allowed_diff last5_oreb_diff rest_diff
```

```
## def_rtg_diff  0              0                      0               0
##              home_back_to_back away_back_to_back shooting_env5 pace_pressure5
## def_rtg_diff  0                 0                 0             0
##              free_throw_env5 tov_env5 missed_fg_pg_diff expected_missed_fg
## def_rtg_diff  0               0        0                 0
##              x3pa_rate_per100_diff last5_missed_fg_diff last10_missed_fg_diff
## def_rtg_diff  0                     0                    0
##              last5_missed_3pt_diff last10_missed_3pt_diff last5_x3pa_rate_diff
## def_rtg_diff  0                     0                      0
##              last10_x3pa_rate_diff last10_oreb_diff last5_oreb_allowed_diff
## def_rtg_diff  0                     0                0
##              last10_oreb_allowed_diff expected_oreb_pressure home_oreb_matchup
## def_rtg_diff  0                        0                      0
##              away_oreb_matchup pace_rebound_pressure
## def_rtg_diff  0                 0
```

Pruned OLS Model 1

```r
pruned_vars <- c(
  "season",
  "rest_diff",
  "home_back_to_back",
  "away_back_to_back",
  "last5_pts_diff",
  "last5_pts_allowed_diff",
  "off_rtg_diff",
  "def_rtg_diff",
  "efg_pct_diff",
  "ft_rate_diff",
  "tov_pct_diff",
  "last5_off_rtg_diff",
  "last5_def_rtg_diff",
  "blend_total_proj5",
  "shooting_env5",
  "pace_pressure5",
  "free_throw_env5",
  "tov_env5",
  "expected_missed_fg"
)

pruned_vars <- pruned_vars[pruned_vars %in% names(X_train_ols)]

lm_pruned <- lm(
  total ~ .,
  data = cbind(total = y_train, X_train_ols[, pruned_vars, drop = FALSE])
)

lm_pruned_pred <- predict(lm_pruned, newdata = X_test_ols[, pruned_vars, drop = FALSE])
lm_pruned_mae  <- mae_raw(y_test, lm_pruned_pred)

cat("OLS Pruned Model - Test MAE:", round(lm_pruned_mae, 2), "\n")
```

```
## OLS Pruned Model - Test MAE: 15.63
```

```r
cat("Pruned R-squared (training):", round(summary(lm_pruned)$r.squared, 4), "\n")
```

```
## Pruned R-squared (training): 0.2727
```

```r
print(summary(lm_pruned))
```

```
##
## Call:
## lm(formula = total ~ ., data = cbind(total = y_train, X_train_ols[,
##     pruned_vars, drop = FALSE]))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -64.994 -12.936  -0.472  12.366 121.529
##
## Coefficients: (1 not defined because of singularities)
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)            -2.439e+03  1.287e+02 -18.954  < 2e-16 ***
## season                  1.180e+00  6.568e-02  17.970  < 2e-16 ***
## rest_diff              -1.336e-02  4.179e-02  -0.320   0.7493
## home_back_to_back      -5.056e-01  4.690e-01  -1.078   0.2810
## away_back_to_back      -6.069e-02  3.874e-01  -0.157   0.8755
## last5_pts_diff         -1.467e-01  7.968e-02  -1.841   0.0657 .
## last5_pts_allowed_diff  9.279e-02  7.923e-02   1.171   0.2416
## off_rtg_diff           -4.296e-02  2.831e-02  -1.518   0.1292
## def_rtg_diff                   NA         NA      NA       NA
## efg_pct_diff            5.848e+00  4.599e+00   1.272   0.2036
## ft_rate_diff            5.240e-01  1.411e+00   0.371   0.7103
## tov_pct_diff            2.265e+00  5.017e+00   0.451   0.6517
## last5_off_rtg_diff      1.653e-01  7.803e-02   2.118   0.0342 *
## last5_def_rtg_diff     -7.666e-02  7.788e-02  -0.984   0.3250
## shooting_env5           1.141e+02  6.863e+00  16.631  < 2e-16 ***
## pace_pressure5          1.988e+00  6.409e-02  31.019  < 2e-16 ***
## free_throw_env5         2.109e+01  4.611e+00   4.573 4.85e-06 ***
## tov_env5               -1.323e+02  1.240e+01 -10.678  < 2e-16 ***
## expected_missed_fg      2.609e-01  6.485e-02   4.023 5.77e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19 on 14470 degrees of freedom
##   (5 observations deleted due to missingness)
## Multiple R-squared:  0.2727, Adjusted R-squared:  0.2718
## F-statistic: 319.1 on 17 and 14470 DF,  p-value: < 2.2e-16
```

OLS on Log-Transformed Total Model 1

```r
lm_log <- lm(
  log(total) ~ .,
  data = cbind(total = y_train, X_train_ols[, pruned_vars, drop = FALSE])
```

```r
)

lm_log_pred_log <- predict(lm_log, newdata = X_test_ols[, pruned_vars, drop = FALSE])
lm_log_pred_raw <- exp(lm_log_pred_log)
lm_log_mae      <- mae_raw(y_test, lm_log_pred_raw)

cat("OLS Log-Transformed - Test MAE (back-transformed):", round(lm_log_mae, 2), "\n")
```

```
## OLS Log-Transformed - Test MAE (back-transformed): 15.67
```

Model 2: Polynomial Regression

```r
poly_base_vars <- intersect(
  c(
    "season",
    "rest_diff",
    "last5_pts_diff",
    "last5_pts_allowed_diff",
    "off_rtg_diff",
    "def_rtg_diff",
    "efg_pct_diff",
    "expected_missed_fg",
    "blend_total_proj5",
    "pace_pressure5"
  ),
  names(X_train_ols)
)

poly_terms <- intersect(
  c("expected_missed_fg", "last5_pts_diff", "last5_pts_allowed_diff"),
  names(X_train_ols)
)

train_poly <- cbind(total = y_train, X_train_ols[, poly_base_vars, drop = FALSE])
test_poly  <- X_test_ols[, poly_base_vars, drop = FALSE]

if ("expected_missed_fg" %in% poly_terms) {
  train_poly <- cbind(
    train_poly,
    poly(X_train_ols$expected_missed_fg, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("missfg_d1", "missfg_d2"))
  )
  test_poly <- cbind(
    test_poly,
    poly(X_test_ols$expected_missed_fg, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("missfg_d1", "missfg_d2"))
  )
}

if ("last5_pts_diff" %in% poly_terms) {
  train_poly <- cbind(
```

```r
    train_poly,
    poly(X_train_ols$last5_pts_diff, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("last5pts_d1", "last5pts_d2"))
  )
  test_poly <- cbind(
    test_poly,
    poly(X_test_ols$last5_pts_diff, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("last5pts_d1", "last5pts_d2"))
  )
}

if ("last5_pts_allowed_diff" %in% poly_terms) {
  train_poly <- cbind(
    train_poly,
    poly(X_train_ols$last5_pts_allowed_diff, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("last5allow_d1", "last5allow_d2"))
  )
  test_poly <- cbind(
    test_poly,
    poly(X_test_ols$last5_pts_allowed_diff, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("last5allow_d1", "last5allow_d2"))
  )
}

lm_poly <- lm(total ~ ., data = train_poly)
lm_poly_pred <- predict(lm_poly, newdata = test_poly)
lm_poly_mae  <- mae_raw(y_test, lm_poly_pred)

cat("Polynomial Regression (degree 2) - Test MAE:", round(lm_poly_mae, 2), "\n")
```

```
## Polynomial Regression (degree 2) - Test MAE: 15.56
```

```r
cat("Polynomial R-squared (training):", round(summary(lm_poly)$r.squared, 4), "\n")
```

```
## Polynomial R-squared (training): 0.2518
```

```r
poly_coefs <- summary(lm_poly)$coefficients
cat("\nSquared-term significance (p-values):\n")
```

```
##
## Squared-term significance (p-values):
```

```r
print(round(poly_coefs[grep("_d2", rownames(poly_coefs)), 4], 4))
```

```
##     missfg_d2   last5pts_d2 last5allow_d2
##        0.6369        0.2361        0.0011
```

Model 3: Gradient Boosted Trees (GBM)

```r
gbm_grid <- expand.grid(
  n.trees           = c(200, 500),
  interaction.depth = c(2, 4),
  shrinkage         = 0.1,
  n.minobsinnode    = 20
)

set.seed(42)
gbm_tune <- train(
  x         = as.data.frame(X_train_gbm),
  y         = y_train,
  method    = "gbm",
  metric    = "MAE",
  trControl = trainControl(method = "cv", number = 3, verboseIter = FALSE),
  tuneGrid  = gbm_grid,
  verbose   = FALSE
)

best_gbm_params <- gbm_tune$bestTune
cat("=== Best GBM Hyperparameters (3-fold CV) ===\n")
```

```
## === Best GBM Hyperparameters (3-fold CV) ===
```

```r
print(best_gbm_params)
```

```
##   n.trees interaction.depth shrinkage n.minobsinnode
## 1     200                 2       0.1             20
```

```r
# Refit on full training set (raw total)
set.seed(42)
gbm_train_df <- as.data.frame(cbind(total = y_train, X_train_gbm))
gbm_test_df  <- as.data.frame(X_test_gbm)

gbm_final <- gbm(
  formula           = total ~ .,
  data              = gbm_train_df,
  distribution      = "gaussian",
  n.trees           = best_gbm_params$n.trees,
  interaction.depth = best_gbm_params$interaction.depth,
  shrinkage         = best_gbm_params$shrinkage,
  n.minobsinnode    = best_gbm_params$n.minobsinnode,
  bag.fraction      = 0.75,
  train.fraction    = 1.0,
  verbose           = FALSE
)

gbm_pred <- predict(
  gbm_final,
  newdata = gbm_test_df,
  n.trees = best_gbm_params$n.trees
)
```

```r
gbm_mae <- mae_raw(y_test, gbm_pred)
cat("GBM Final Model - Test MAE:", round(gbm_mae, 2), "\n")
```

```
## GBM Final Model - Test MAE: 15.49
```

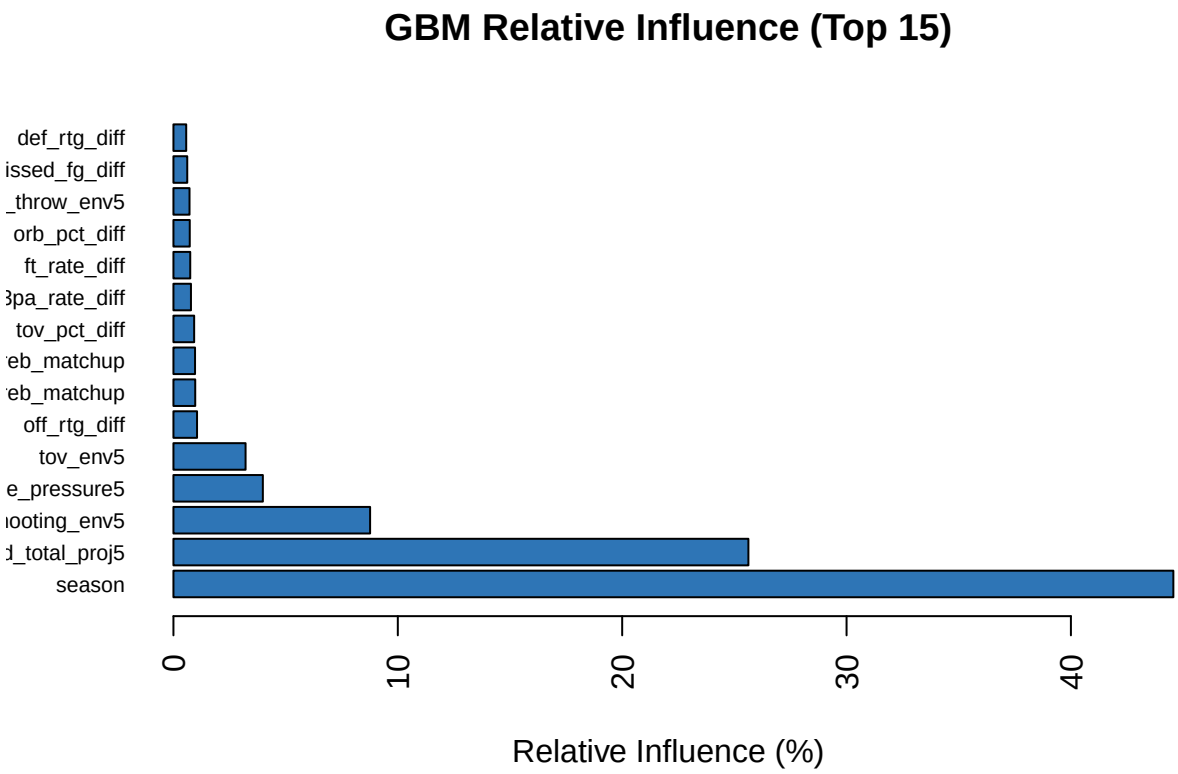
```r
# Variable importance
cat("\nGBM - Variable Importance (top 15):\n")
```

```
##
## GBM - Variable Importance (top 15):
```

```r
gbm_summary <- summary(
  gbm_final,
  n.trees = best_gbm_params$n.trees,
  plotit = FALSE
)
print(head(gbm_summary, 15))
```

```
##                                          var    rel.inf
## season                                season 44.5656874
## blend_total_proj5         blend_total_proj5 25.6261005
## shooting_env5                 shooting_env5  8.7649420
## pace_pressure5               pace_pressure5  3.9847333
## tov_env5                           tov_env5  3.2153414
## off_rtg_diff                   off_rtg_diff  1.0514447
## home_oreb_matchup         home_oreb_matchup  0.9704374
## away_oreb_matchup         away_oreb_matchup  0.9628701
## tov_pct_diff                   tov_pct_diff  0.9231317
## last10_x3pa_rate_diff last10_x3pa_rate_diff  0.7807345
## ft_rate_diff                   ft_rate_diff  0.7489339
## orb_pct_diff                   orb_pct_diff  0.7269312
## free_throw_env5             free_throw_env5  0.7162166
## last5_missed_fg_diff   last5_missed_fg_diff  0.6157206
## def_rtg_diff                   def_rtg_diff  0.5688162
```

```r
barplot(
  head(gbm_summary$rel.inf, 15),
  names.arg = head(gbm_summary$var, 15),
  horiz     = TRUE,
  las       = 2,
  main      = "GBM Relative Influence (Top 15)",
  xlab      = "Relative Influence (%)",
  col       = "#2E75B6",
  cex.names = 0.7
)
```




```r
# Log-scale GBM
set.seed(42)
gbm_log_train_df <- as.data.frame(cbind(total_log = y_train_log, X_train_gbm))
gbm_log_test_df  <- as.data.frame(X_test_gbm)

gbm_log_final <- gbm(
  formula           = total_log ~ .,
  data              = gbm_log_train_df,
  distribution      = "gaussian",
  n.trees           = best_gbm_params$n.trees,
  interaction.depth = best_gbm_params$interaction.depth,
  shrinkage         = best_gbm_params$shrinkage,
  n.minobsinnode    = best_gbm_params$n.minobsinnode,
  bag.fraction      = 0.75,
  train.fraction    = 1.0,
  verbose           = FALSE
)

gbm_log_pred <- exp(
  predict(
    gbm_log_final,
    newdata = gbm_log_test_df,
    n.trees = best_gbm_params$n.trees
  )
)
```

```r
gbm_log_mae <- mae_raw(y_test, gbm_log_pred)
cat("GBM (log scale, back-transformed) - Test MAE:", round(gbm_log_mae, 2), "\n")
```

```
## GBM (log scale, back-transformed) - Test MAE: 15.57
```